

Ben Irastorza

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SUMMARY

Software Engineer with strong programming, mathematics, and data analysis skills. Experienced in developing AI and data-driven applications, with a background in conducting in-depth research.

EDUCATION

Georgia Institute of Technology

Atlanta, GA

Bachelor of Science in Computer Science

May 2026

GPA: 3.5

- **Concentration:** Artificial Intelligence and Information Internetworks
- **Relevant Coursework:** Data Structures & Algorithms, Systems & Networks, Computer Organization & Programming, Database Systems, Linear Algebra, Multivariable Calculus, Probability & Statistics
- **Involvement:** IFC Social Committee Lead Safety Coordinator, Lambda Chi Alpha Fraternity Founding Father, Catholic Center

TECHNICAL SKILLS

Languages: Python, Java, C, SQL

Libraries/Frameworks: LangChain, NumPy, Pandas, JUnit, JavaFX

Development Tools: Jira, Git, Docker, Mamba, Figma, Drawio

Cloud/Databases: PostgreSQL, MySQL, Firebase, Milvus

EXPERIENCE

Undergraduate Researcher

Aug 2024 – Present

Georgia Tech Automotive LiDAR Lab

Atlanta, GA

- Researching advanced automotive systems to enhance vehicle autonomy through LiDAR technology.
- Developing Python-based edge detection algorithms using OpenCV in a containerized ROS2 environment.
- Collaborating with the electrical team to test and optimize sensor data for enhanced driving performance.

Data Science Intern

May 2024 – Aug 2024

The Parker Avery Group

Atlanta, GA

- Built the firm's Generative AI initiatives within an Agile (Scrum) framework, applying retail analytics knowledge.
- Achieved a 50% reduction in AI processing costs and improved efficiency by leveraging the Batch API to enable asynchronous processing of large-scale data jobs.
- Increased predictive model accuracy by 20%+ by generating a specialized training dataset using advanced PostgreSQL queries to address forecast bias and accuracy queries.
- Ensured consistent setups and streamlined onboarding by standardizing development environments with Mamba YAML files, supported by comprehensive and well-structured documentation.

PROJECTS

Enhanced Spotify Wrapped | *Java, Android SDK, Firebase, Git, Figma*

- Integrated custom AI to enhance the app with conversational capabilities tailored to individual user preferences, personalizing and improving the app experience.
- Implemented a CI/CD pipeline using GitHub Actions for automating testing with Gradle, reducing build errors.
- Used Firebase to manage account creation, login, and secure user credentials, ensuring robust data protection.
- Designed the front end in Figma and transferred the layouts to XML for Android implementation, ensuring design consistency across the app.

Personal Computer Building Business

- Oversee 10 client projects, delivering custom-built systems with a 100% satisfaction rate through precise component selection, assembly, and ongoing support.
- Evaluate CPU core count and architecture to match workload characteristics, prioritizing IPC for single-threaded or parallelism for multithreaded tasks.

Portfolio Website | *HTML, CSS, Git*

- Building a responsive portfolio website to overview my skills, projects, and experience comprehensively.
- Implementing HTML for structured content while leveraging CSS for visual styling and layout design.